Leukemia Blood Cells

Sorting Blood Cells With AI — and Why 99.8% Made Me Suspicious

Capstone Project Report

A model that sorts leukemia cells from blood-smear photos — plus a lesson in why an almost-perfect score can actually be a warning sign.

RL · RLcapstone.ai

1. Summary

This project looks at a photo of one white blood cell and sorts it into four types: a healthy look-alike, plus three stages of a blood cancer called ALL (acute lymphoblastic leukemia). It scored **99.8%** on the test photos.

> I'm not showing off that 99.8% — I'm **suspicious** of it. This dataset doesn't tell you which patient each cell came from, so cells from the same patient probably ended up in both training and testing. That means the model might be winning by recognizing stuff like the exact staining color, not actual cancer signs.

2. Why I built this

Telling a healthy look-alike cell from a real cancer cell — and telling the cancer stages apart — is genuinely hard even for experts, so it's a cool image problem. It's also the most serious topic I worked on, so being honest about the results mattered the most here.

3. The data

I used a public dataset (from Kaggle) of 3,256 blood-smear photos, each showing one cell, across the four types. I split them 70/15/15 into training / checking / testing.

The problem I couldn't fully fix

The dataset doesn't include patient IDs (the file names are just numbers), so there's **no way to guarantee** that one patient's cells stay together. That means some 'cheating' is probably baked in — and pretending it isn't would be the dishonest thing to do.

4. How it works

It's the same image AI I used for MoleCheck (YOLO11), just retrained on blood cells. It hit 100% on the checking set within about 30 rounds of training — which, again, is a little *too* good.

5. Results

What I measured	Score
Test accuracy	99.8%
Fair score across types	0.998
Mistakes	1 out of 489 photos

On most projects this would be the big headline. Here it's the red flag. Cells from the same slide share the same coloring, lighting, and background, and a strong model can score almost perfectly just by picking up on **those** instead of the actual cell.

6. Getting it running

I put it in a web demo (pick a real cell photo, sort it on your device) and an Android app, both matching the original model.

7. Honest limitations

- Not a diagnosis tool — real leukemia diagnosis needs lab tests like bone-marrow analysis.
- The 99.8% is probably inflated by the data problem above, so treat it as a best case, not reality.
- It's one specific dataset; real hospitals see way more variety.

8. What's next

The clear next step is a bigger dataset called C-NMC that **does** include patient IDs, so I can keep each patient's cells together (the way I did for the brain-wave project). The score will probably drop — but a lower, honest number is worth way more than this one.